ENDOVASCULAR GRAFT SYSTEMS AND METHODS FOR DEPLOYMENT IN MAIN AND BRANCH ARTERIES

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] ~~This application claims priority to U.S. Provisional Patent Application No. 62/529,669, filed on July 7, 2017, which is hereby incorporated by reference in its entirety.~~

FIELD

[0002] Arrangements disclosed herein relate generally to the field of endovascular graft systems. In particular, various arrangements relate to endovascular graft systems for placement in a thoracic endoluminal space and allowing for the perfusion of blood to the greater arch vessels. Various arrangements also relate to endovascular graft systems with kink resistance features.

BACKGROUND

[0003] The present disclosure relates to systems and methods for treating vasculature disorders, such as aneurysms. An aneurysm is a medical condition characterized by an expansion and corresponding weakening of the wall of an artery of a person. Aneurysms can develop at different sites within an artery. For example, thoracic aortic aneurysms or abdominal aortic aneurysms may manifest within a person's body. This medical condition is serious and potentially life threatening to the person, thereby requiring medical intervention to treat the condition. Existing systems and methods for treating such conditions include invasive surgical procedures with graft replacement of the affected vessel or body lumen or reinforcement of the vessel with a graft.

[0004] Surgical procedures to treat aneurysms are associated with relatively high morbidity and mortality rates due to risk factors inherent to surgical repair of an artery or artery wall. Long and painful recoveries are often required, resulting in extensive medical costs. Because of the inherent risks and complexities of surgical repair of an aortic aneurysm, endovascular repair has become a widely used alternative therapy. US 2005/0228484 Al discloses a modular endovascular graft.

SUMMARY OF THE DISCLOSURE

[0005] The invention is defined in claim 1. Further aspects and preferred embodiments are defined in the dependent claims. Any aspects, embodiments and examples of the present disclosure which do not fall under the scope of the appended claims do not form part of the invention and are merely provided for illustrative purposes. Various arrangements disclosed herein are directed to systems and methods for treating diseased bodily lumens involving branched lumen deployment sites. Various methods include using the systems disclosed herein to maintain blood flow through a main artery and from the main artery to one or more branch arteries.

[0006] In various arrangements, an endovascular graft or section thereof includes sealing rings with stent segments interspersed in-between the sealing rings. In various arrangements, an endovascular graft system for deployment at branched arteries includes a main tubular graft body deployable within a main artery comprising a proximal end and an opposed distal end, where the proximal and distal ends define a tubular graft wall therein between. In various arrangements, the endovascular graft system includes a plurality of inflatable channels disposed along the main tubular graft body, and ~~at least one~~a plurality of stent segments that are~~is~~ disposed along the tubular graft wall of the main tubular graft body. ~~In some arrangements, t~~The plurality of inflatable channels are configured to be inflatable with an inflation medium, and ~~the at least one~~a plurality of stent segments are ~~is~~ disposed between two or more adjacent inflatable channels.

[0007] In some arrangements, the at least one stent segment is at least one ring segment. In some arrangements, the at least one stent segment has a C-shape. In some arrangements, the at least one stent segment is at least one nested ring segment. In some arrangements, the at least one stent segment is a continuous wire wound stent. In some arrangements, the at least one stent segment is a helical continuous wire wound stent. In some arrangements, the at least one stent segment has a circular cross sectional shape. In some arrangements, the at least one stent segment has a partially circumferential cross sectional shape. In some arrangements, the at least one stent segment has a completely circumferential cross sectional shape.

[0008] In various arrangements, the main tubular graft body has at least one fenestration disposed between two or more adjacent inflatable channels of the plurality of inflatable channels. In some arrangements, the at least one fenestration is configured to be in fluid contact with at least one branch artery branched from a main artery. In some arrangements,

[0093] An endovascular graft or section thereof in accordance with various arrangements includes sealing rings with stent segments interspersed in-between the sealing rings. In various arrangements, the sealing rings are polymer sealing rings and are configured to hold a blood flow lumen open when filled with polymer. In some arrangements, an interstitial space between adjacent sealing rings includes a graft. In some arrangements, the graft is made of Polytetrafluoroethylene (PTFE) or the like. In various arrangements, the endovascular graft is deployable in angulated anatomies.

[0094] In various arrangements, a stent is interspersed in-between polymer sealing rings. In some arrangements, the stent is a continuous helical wire wound stent. In some arrangements, the stent comprises multiple series of individually nested stent rings. In various arrangements, the stent provides support to keep a lumen open while the endovascular graft is curved. In various arrangements, the stent structure is intimately laminated into the graft. In some arrangements, the stent structure is adhered to or bonded to the graft by other means, such as by sutures, glue, or the like. In various arrangements, the stent material is made from Nitinol or other suitable material. In various arrangements, the stent is made of wire or laser cut from tubing and then expanded to a suitable diameter. In some arrangements, stent segments are interspersed in-between polymer sealing rings.

[0095] FIG. 1 shows an endovascular system in accordance with an arrangement. An endovascular system in accordance with an arrangement includes a main tubular graft body 100 which includes wall portions 110 disposed between a plurality of inflatable channels 106. ~~One or more~~A plurality of stent segments 104 are disposed between two or more of the inflatable channels 106. In some examples, two of the wall portions 110 are connected by one of the inflatable channels 106. ~~In various arrangements, e~~Each stent segment 104 is disposed on a surface of a corresponding wall portion 110 or laminated into the wall portion 110. As such, each stent segment 104 contacts or is embedded in the surface of a corresponding wall portion 110. The main tubular graft body 100 also includes wall portions 111 at both a proximal end of the main tubular graft body 100 and a distal end of the main tubular graft body 100. In some examples, the wall portions 111 do not have any stent segments 104 disposed thereon.

[0096] The main tubular graft body 100 also includes anchor members 102 disposed at both the distal end and the proximal end of the main tubular graft body 100. The anchor members

the plurality of inflatable channels 106 shown in FIG. 2 are configured as annular rings instead of helical channels.

[0099] ~~In various arrangements, t~~The inflatable channels 106 are annular rings and the stent segments 104 are segmented stent rings. In various arrangements, the annular inflatable channels 106 are non-helical. In some arrangements, the stent segments 104 are discrete, separate, and segmented stent rings nested in interstitial spaces between two adjacent inflatable channels 106. In various arrangements, the stent segments 104 (e.g., stent ring segments) are completely circumferential (O-shaped, around the main tubular graft body 100) around a circumference of the main tubular graft body 100. In some arrangements, the stent segments 104 are partially circumferential (such as being C-shaped instead of O-shaped). In various arrangements, the stent segments 104 (e.g., nested stent ring segments) are made in continuous segments between the inflatable channels 106 (e.g., annular rings) such that a zig pattern of the stent segments 104 can jump to an adjacent zig pattern within the inter-ring segment (e.g., a same wall portion 110). A resultant wire spine for the stent segments 104 can be aligned with a polymer spine fill channel (not shown) configured to fill the inflatable channels 106 (e.g., annular rings).

[0100] Referring to FIG. 3A, a typical configuration of an aorta 10 of a person is shown. The aorta 10 includes a descending aorta portion 11 (i.e., a distal aortic portion), an aortic arch portion 12, and an ascending aorta portion 13 (i.e., a proximal aortic portion). The aorta 10 is fluidically connected to the left subclavian artery 14, the left common carotid artery 15, the right common carotid artery 16, the right subclavian artery 17, the right coronary artery 18, and the left coronary artery 19. The great vessels arise off the aortic arch portion 12 and include the innominate artery (also known as the brachiocephalic artery) 20, the left common carotid artery 15, and the left subclavian artery 14, which are referred to as the aortic arch vessels. The innominate artery (also known as the brachiocephalic artery) 20 divides into the right common carotid artery 16 and the right subclavian artery 17.

[0101] FIG. 3B shows an endovascular system of the present disclosure in accordance with an arrangement as deployed in an aorta 10 of a typical person. An endovascular system in accordance with various arrangements as shown in FIG. 3B has a fenestration 114 and includes a branched device 300 for insertion into the left subclavian artery 14. The